Required Graphs for Adaptive AI Interview System – TabPFN Research

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PROJECT CONTEXT
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Project: Adaptive AI Interview System

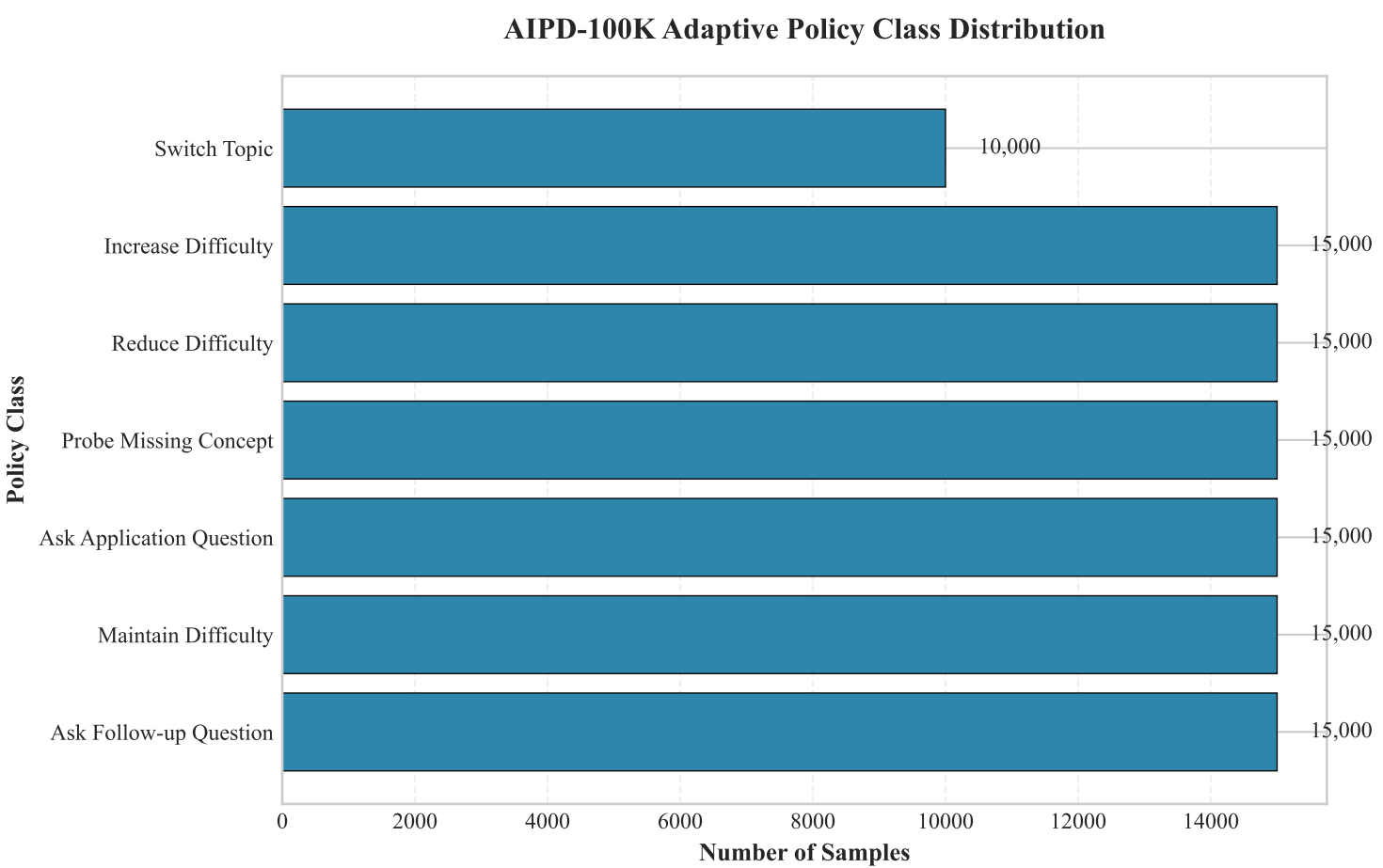
Primary research objective:
Develop an adaptive interview system that analyzes the candidate's current interview state and selects the next interview policy.

Primary model: TabPFN
Baseline models: Random Forest, XGBoost, CatBoost
Dataset: AIPD-100K (100,000 samples, 11 features, 7 policy classes)

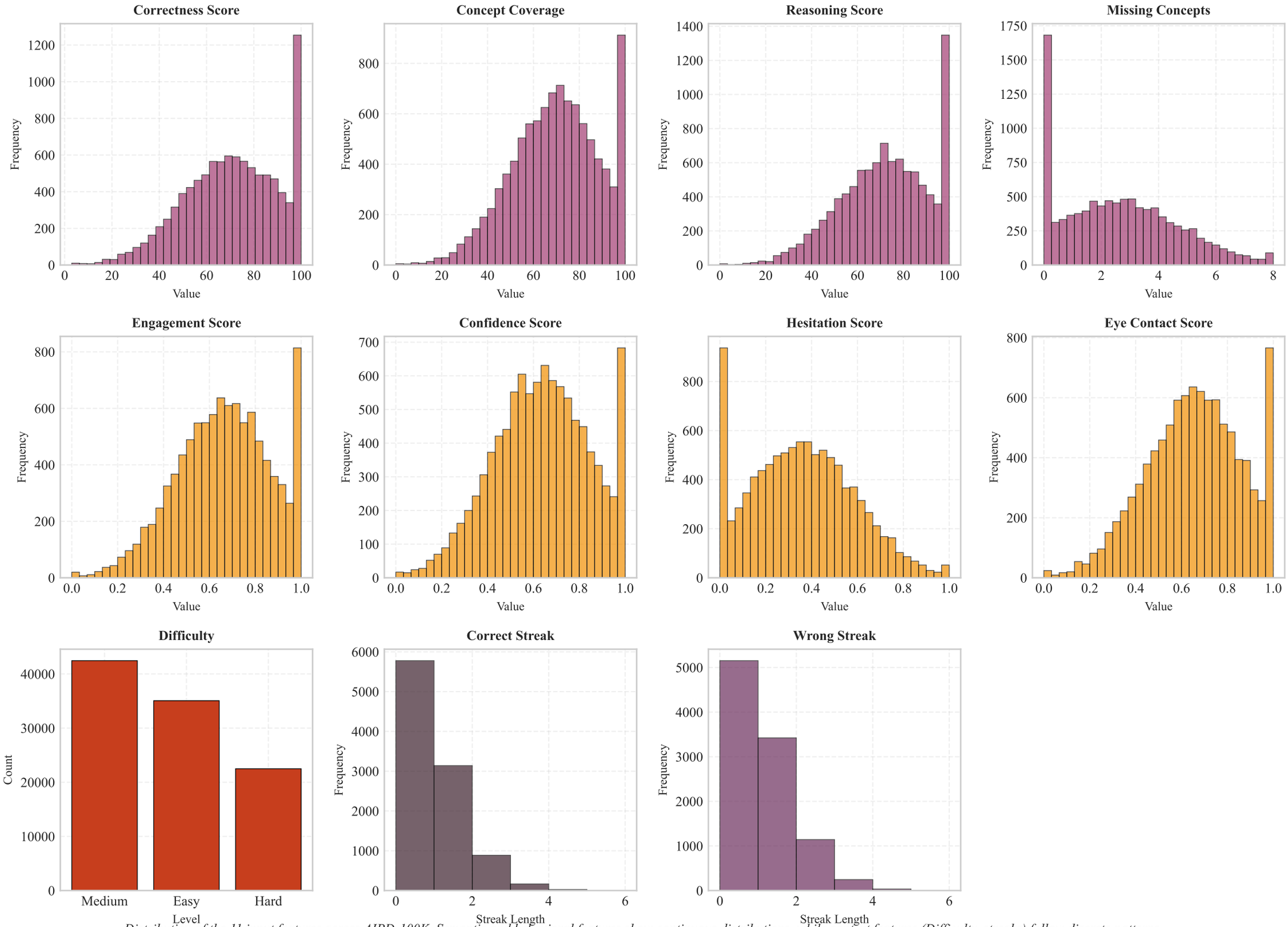
GRAPH REQUIREMENTS
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This document contains 7 professional graphs required for the IEEE research paper:

1. Dataset Target Class Distribution
2. Feature Distributions
3. Model Performance Comparison
4. TabPFN Training-Size Scalability
5. Training Size vs Inference Time
6. TabPFN Confusion Matrix
7. Behavioral Feature Derivation

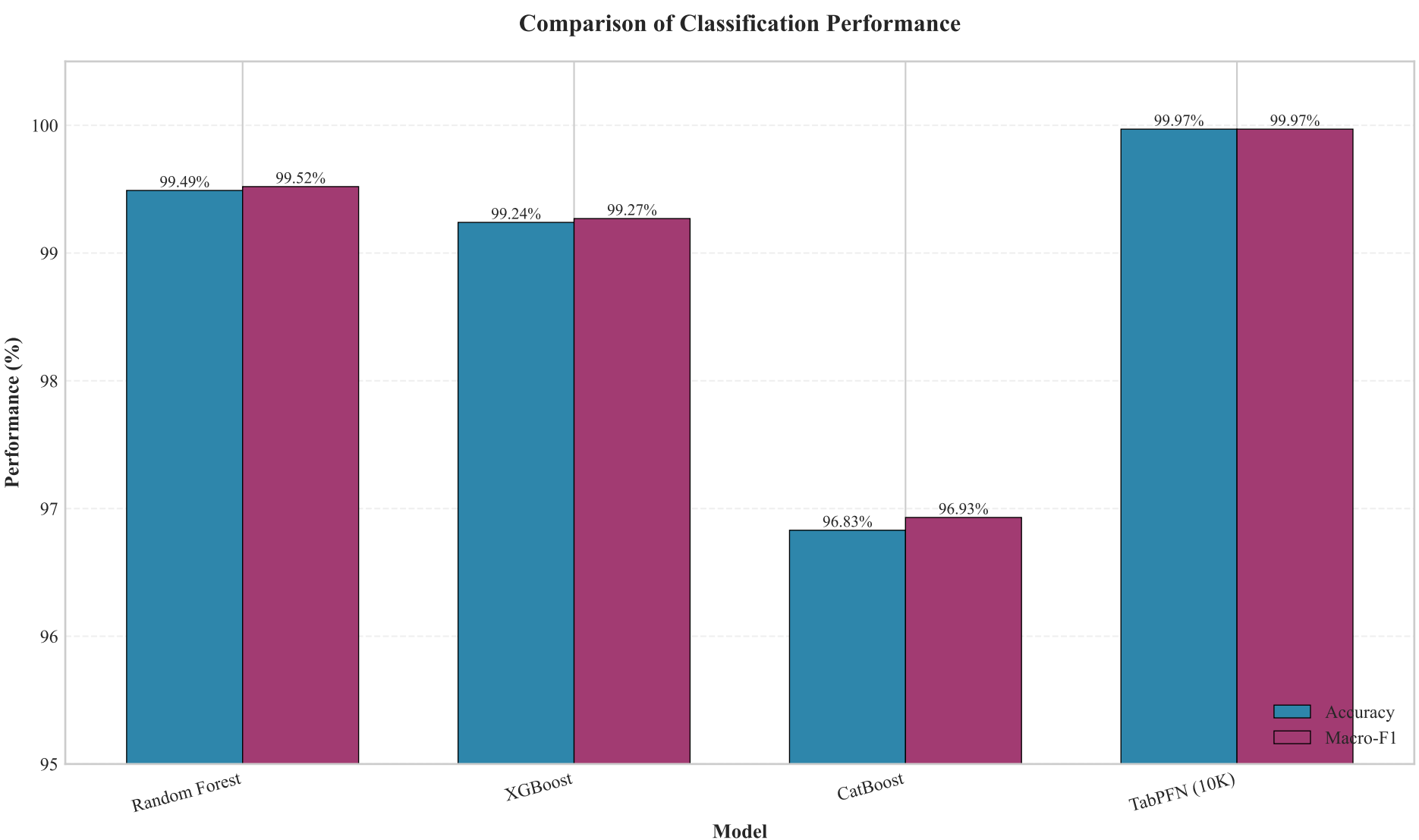
All graphs use actual experimental data from the project. No values are fabricated.
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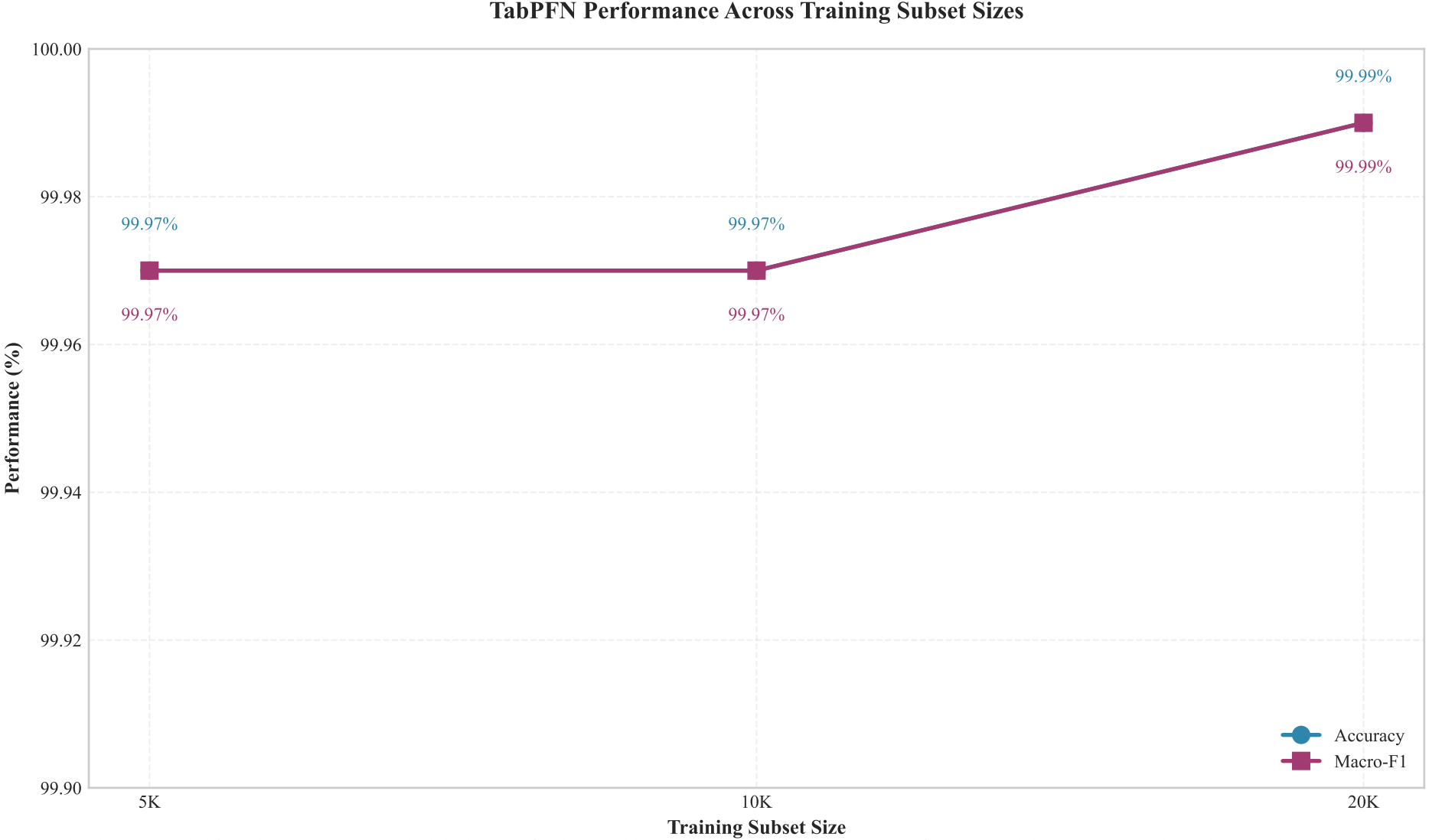
Distribution of the 7 adaptive interview policy classes in AIPD-100K. The dataset is well-balanced with each policy class represented by 10,000-15,000 samples.



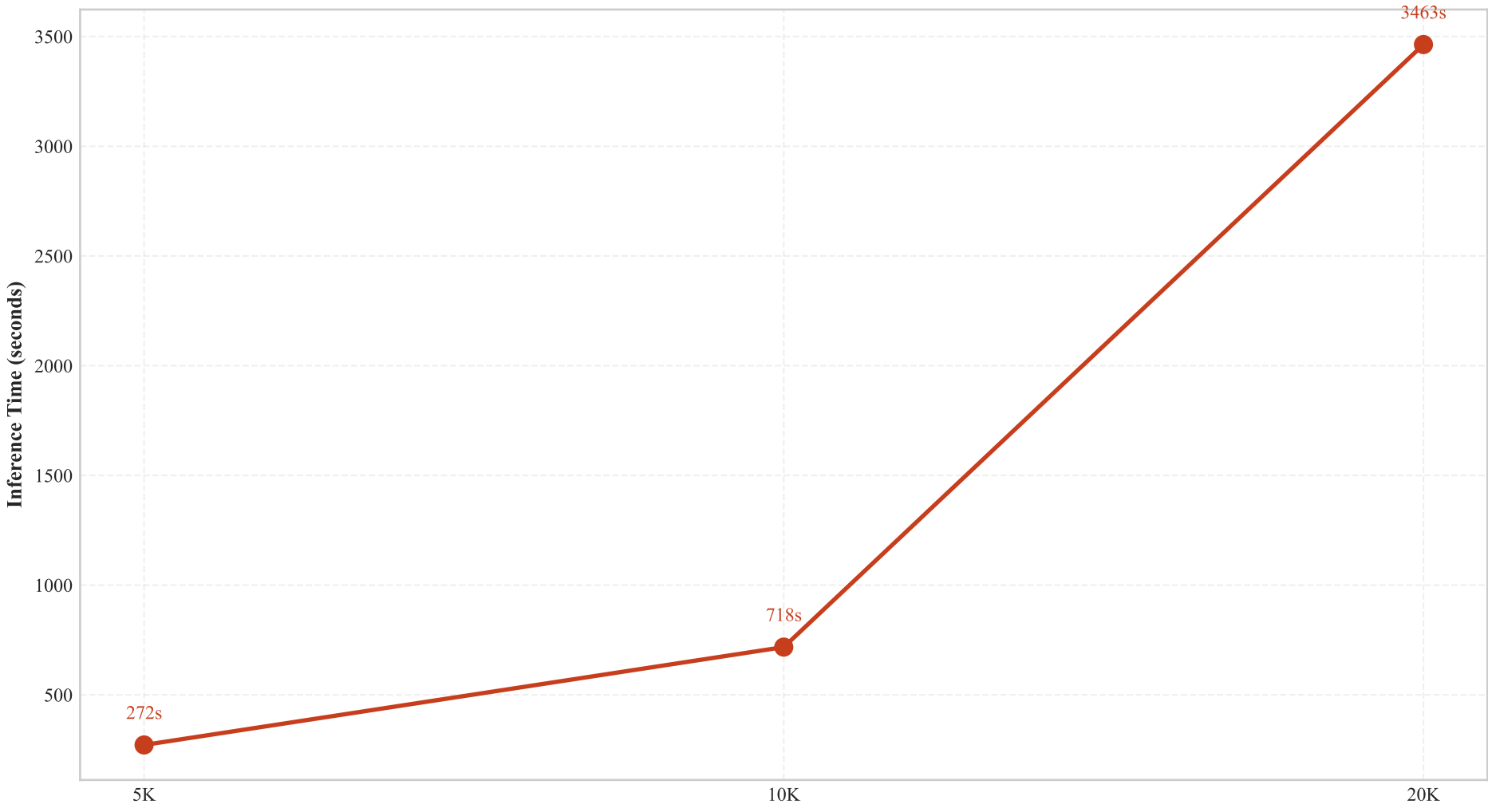
Distribution of the 11 input features across AIPD-100K. Semantic and behavioral features show continuous distributions, while context features (Difficulty, streaks) follow discrete patterns.



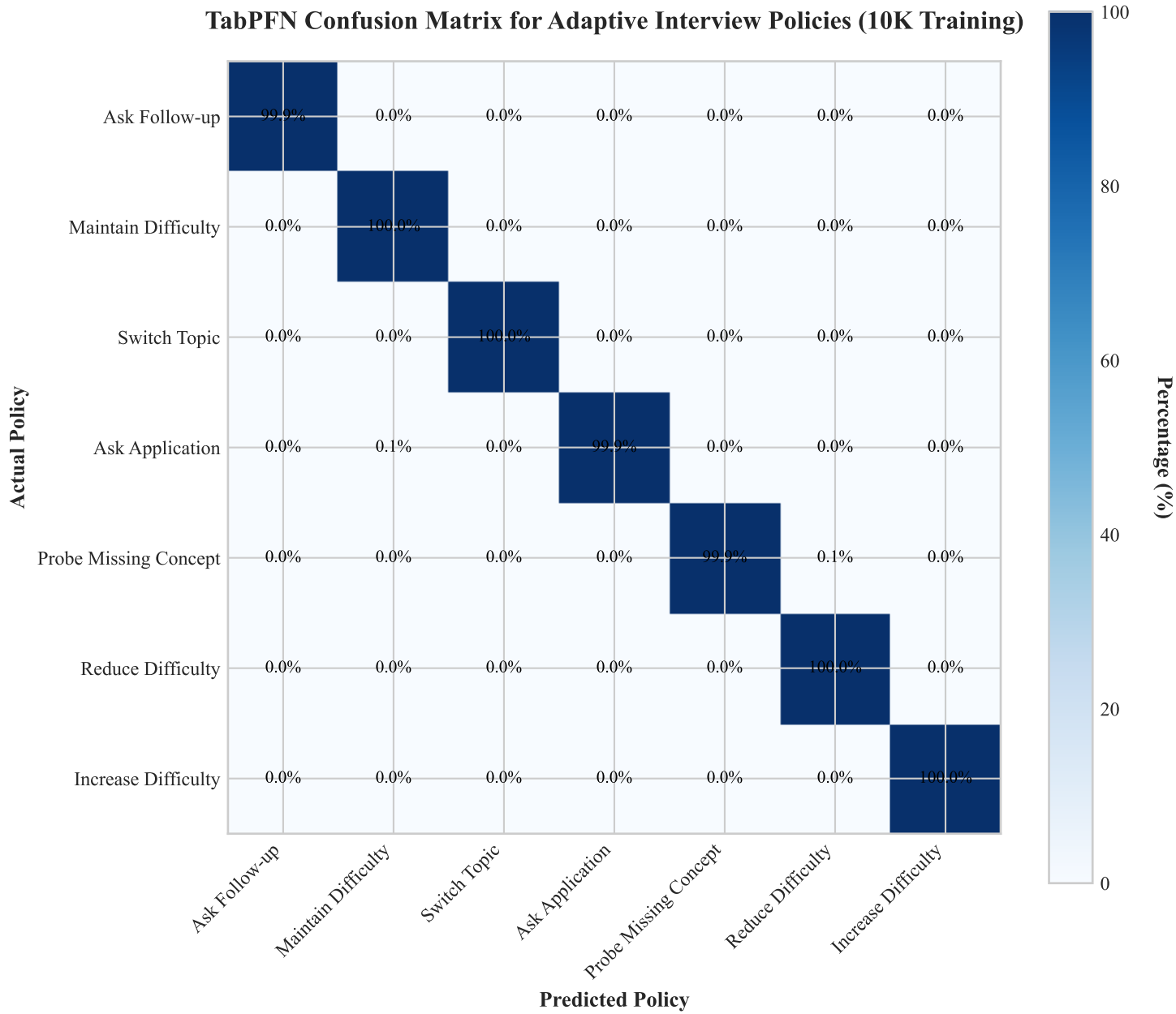
Comparison of accuracy and macro-F1 across conventional baseline models (Random Forest, XGBoost, CatBoost) and the proposed TabPFN approach (10K training samples).



Each subset is sampled from the 70K training pool while the same fixed 15K held-out test set is used for evaluation. Performance stabilizes at 10K samples with minimal gains beyond.

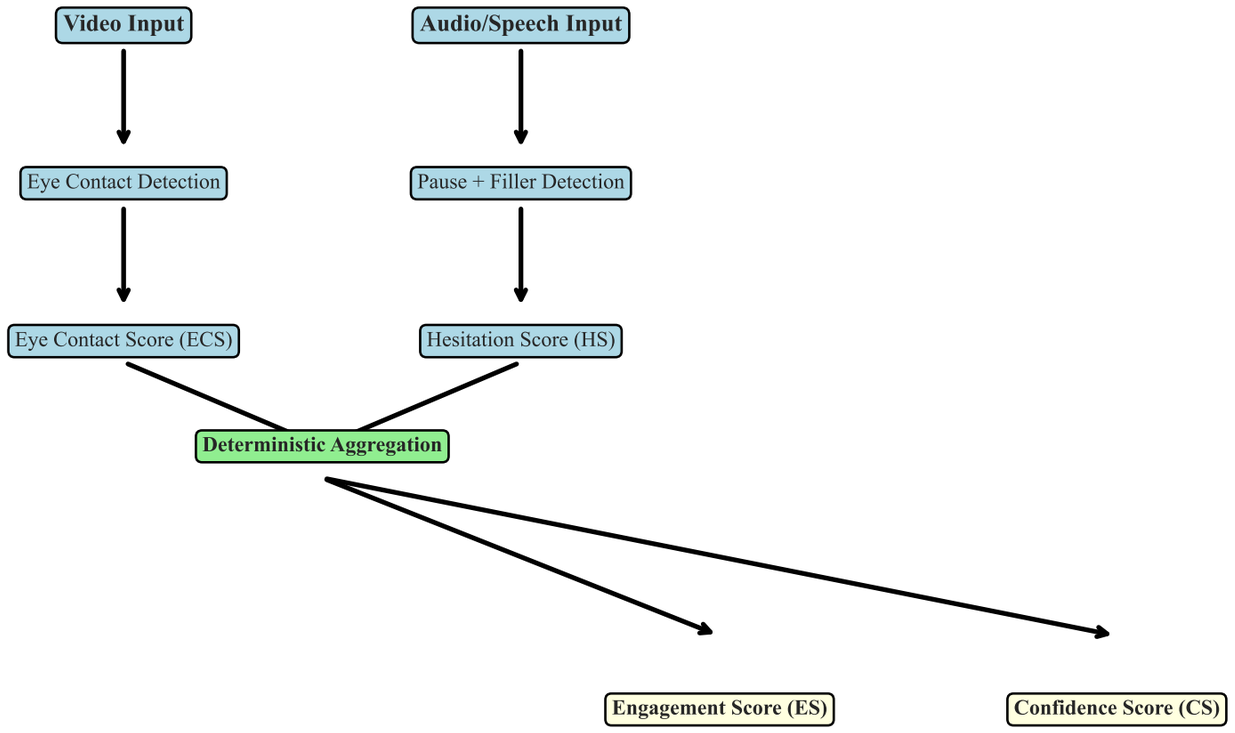


Inference time increases with the number of task-specific training examples, illustrating the computational trade-off associated with larger training subsets due to transformer attention complexity.



Confusion matrix for TabPFN (10K training samples) showing near-perfect classification across all 7 adaptive interview policies. Diagonal dominance indicates excellent model performance.

Behavioral Feature Derivation Pipeline



$$ECS = (T_{eye}/T_{valid}) \times 100 \qquad HS = 100 \times (0.7P + 0.3F)$$

$$ES = 0.6(ECS) + 0.4(100 - HS) \; CS = 0.5(ECS) + 0.5(100 - HS)$$

Process diagram showing how raw video and audio signals are transformed into the four behavioral features (Eye Contact Score, Hesitation Score, Engagement Score, Confidence Score) used by TabPFN. These are derived behavioral proxy scores, not direct measurements of psychological states.

Recommended Graphs for the IEEE Research Paper

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ESSENTIAL GRAPHS (Main Paper)
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These graphs should definitely appear in the main paper:

1. Model Performance Comparison (Figure 3)
   - Establishes TabPFN performance against baselines
   - Critical for research contribution validation
   - Position: Results section, primary performance comparison

2. TabPFN Training-Size Scalability (Figure 4)
   - Demonstrates few-shot learning capability
   - Shows performance stabilization point
   - Position: Results section, scalability analysis

3. TabPFN Confusion Matrix (Figure 6)
   - Shows class-wise prediction performance
   - Validates no policy confusion
   - Position: Results section, detailed performance analysis

4. Training Size vs Inference Time (Figure 5)
   - Shows computational trade-offs
   - Important for practical deployment discussion
   - Position: Results section or Discussion, computational analysis

SUPPORTING GRAPHS (Supplementary Material)
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These graphs can be moved to supplementary material if page space is limited:

5. Dataset Target Class Distribution (Figure 1)
   - Shows dataset composition
   - Can be mentioned in text without full figure
   - Position: Methods section or supplementary

6. Feature Distributions (Figure 2)
   - Shows feature characteristics
   - Can be summarized in text description
   - Position: Methods section or supplementary

7. Behavioral Feature Derivation (Figure 7)
   - Explains behavioral feature pipeline
   - Can be described in text with simple diagram
   - Position: Methods section or supplementary

RECOMMENDATION
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For a standard IEEE conference paper (6-8 pages), include all 4 ESSENTIAL graphs in the main paper.
Move the 3 SUPPORTING graphs to supplementary material if space is constrained.

The Model Performance Comparison and TabPFN Scalability graphs are the most critical for establishing the research contribution.
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